

Dr. rer. nat. Julia Thomas

Lead Machine Learning Engineer · MLOps & Data Engineering

Berlin · julia@xemantic.com

PROFILE

Machine learning engineer owning the entire ML lifecycle – from cloud infrastructure and data pipelines through LLM platforms to autonomous, neurosymbolic agents. Trains and deploys custom models (from classical ML and Graph Neural Networks through transformers to agentic LLM systems) and combines deep MLOps and data-engineering expertise with a rigorous, methods-and-statistics ML foundation (doctorate in clinical psychology & epidemiology). Focus areas: mechanistic interpretability, recalibration, and the realistic validation of model capabilities beyond standard benchmarks – not merely wiring up LLM APIs. Strong interest in efficient, resource-conscious, and ethically sound AI applications.

PROFESSIONAL EXPERIENCE

Rulemapping Solutions, Berlin

August 2025 – Present

Lead Machine Learning, MLOps & Data Engineer

Neurosymbolic AI for legal tech and regulatory compliance. Steering and strategic direction of the AI department – roadmap, directional decisions, and hands-on delivery across the full ML lifecycle as an end-to-end stack:

- **Cloud Infrastructure:** built the entire cloud infrastructure with Terraform, NixOS, and Kubernetes – reproducible, scalable, and resilient
- **Deployment & Operations:** rollout of all services via build and deploy pipelines in multiple formats (DEB/RPM packages, Docker, Kubernetes, OpenShift); resilient production operation with observability, fallback, and incident mechanisms
- **Data Engineering & Integration:** standardized ingestion from heterogeneous sources and custom data-driven microservices (e.g. geodata/maps, the German special electronic attorney mailbox); preparation of difficult document structures (OCR, chunking, standardization) for reliable LLM processing
- **LLM Middleware (Unified Proxy, self-built):** self-implemented interface that unifies access across providers and LLM API formats (IBM Watson, Azure, Ollama); built-in observability plus add-on capabilities that can be enabled modularly and uniformly per project – RAG, OCR pipelines, chunking, tool use – including MLflow tracing
- **Workload Optimization & Model Adaptation:** improved time and token efficiency of LLM-based workloads in neurosymbolic decision graphs (parallelization, budgeting, chunking); retraining, recalibration, instruction fine-tuning, and mechanistic interpretability to adapt models to client data – with attention to the models' probabilistic properties
- **Neurosymbolic Chatbot:** chat solution on top of the in-house XML-based framework, with prompt scaffolding and integration of the knowledge graph
- **Agentic Knowledge Graph (development focus):** self-maintaining, continuously self-enriching graph as the central component – autonomous chain checks, automatic triggering of data pipelines, evaluation within the conversational context (Graph-RAG); currently extending it with deep graph learning toward a hybrid approach
- **Observability & MLflow:** built observability measures and infrastructure covering both LLM workloads and the training of classical models (MLflow tracing, monitoring, traceability of decisions)
- **Prompt Engineering & Guardrailing:** support legal engineers with prompt engineering and empirical prompt tuning; built pipelines with guardrailing to ensure safety, consistency, and stability of outputs
- **AI Department Lead:** steering and strategic direction of the department – roadmap, directional and optimization decisions, and their hands-on implementation; conception of the knowledge graph

Moodbase e.G. i.G., Berlin

March 2025 – Present

Founding Engineer

Provision of AI- and enterprise-ready data on the heterogeneous field of psychosocial care – plus a free public platform for people seeking help.

- Built a fully agentic data-aggregation and scraping pipeline that maps the care landscape – automated collection and refresh of data from the web via self-steering, self-improving update mechanisms with an adaptive search process
- Delivery of curated, current, enterprise-ready datasets for municipalities and other clients that want to enrich their own offerings (e.g. recommender systems and chatbots) with well-maintained data

- Operation of a free directory and exchange platform that lists and categorizes services for people seeking help
- Enrichment of data in multiple formats and in knowledge graphs; chatbot solutions that consolidate information across the graph (Graph-RAG)
- Responsible for platform development, data architecture, AI tooling, and technical infrastructure

Krisenchat gGmbH, Berlin

June 2021 – October 2025

Germany's largest crisis text line for young people – AI assistance systems supporting counselors while preserving therapeutic agency and ensuring GDPR compliance.

Senior AI Solutions Architect

January 2025 – October 2025

- Architected AI systems that support counselors without limiting therapeutic agency; bridged therapeutic requirements and technical implementation
- Technical lead for AI decisions; designed secure, GDPR-compliant infrastructure for sensitive health data
- Identified and mitigated risks around data privacy, therapeutic safety, and AI ethics in hybrid counseling contexts

Machine Learning Engineer & Data Engineer

November 2022 – October 2025

AI systems & custom models:

- Trained numerous custom deep-learning models – from classical ML through transformers to agentic LLM applications
- Multivariate time-series transformer for capacity planning and waitlist prioritization
- Dual-encoder sentence-transformer for fine-grained risk detection (suicide-risk monitoring)
- Heterogeneous Graph Neural Network for complex risk and trajectory prediction (deep graph learning)
- RAG tools, recommender and matching systems for therapists – including matching of special cases
- Agentic LLM pipelines and chatbots to plan complex counseling strategies; deployed custom LLMs for automated documentation and counseling support
- Fully autonomous LLM applications performing self-directed diagnostic workloads (end-to-end)

MLOps & infrastructure:

- Designed and operated an automated MLOps architecture for the full model lifecycle – from MLflow experimentation to deployment on GCP
- CI/CD pipelines for continuous model development; automated ETL workflows; cloud engineering with Terraform for scalable ML infrastructure

Research Scientist

June 2021 – November 2022

- Service evaluations, analytics, and dashboard development for decision support
- Data-warehousing architecture and ETL workflows for data integration; first AI models for risk management
- Supervised master's students; led publication processes from analysis to dissemination

Ethical Technologies for Mental Health UG, Berlin

August 2024 – Present

Founding AI Engineer

- Conception and implementation of a knowledge-graph interface for automatic data ingestion
- Development of ML pipelines for graph agglomerations
- LLM-based data retrieval from the knowledge graph

Private Psychotherapy Practice, Berlin

Since 2025

Psychotherapist in Training (PiA)

- Psychotherapeutic treatment of own patients in individual settings (under supervision)
- Application of evidence-based behavioral-therapy methods

Klinik Bergfried, Saalfeld

October 2024 – August 2025

Clinical Psychologist / Psychotherapist in Training

- Individual and group psychotherapy in a rehabilitation setting
- Competency training and relaxation techniques
- Clinical experience informing human-centered AI design

Xemantic, Berlin

February 2021 – Present

Co-founder – applied-philosophy & AI research collective

Independent AI research lab and computational-art studio at the intersection of applied philosophy and code.

- Research and development on Golem XIV, an autonomous, metacognitive AI system with semantic memory and self-directed research
- Self-managed knowledge-graph memory (Neo4j) with cognitive-process observability for human oversight
- Chain-of-code reasoning in domain-specific languages and metacognitive analysis capabilities

International Psychoanalytic University, Berlin

December 2019 – October 2021

Quality Management & Organizational Development

- Statistical analysis and evaluation of university programs; qualitative assessments
- Designed new evaluation formats; developed and implemented monitoring databases
- Scientific reporting and presentation of results

Deutsche Kinder- und Jugendstiftung gGmbH, Berlin

November 2018 – December 2019

Working Student – Evaluation & Development

- Inferential statistical analysis; evaluation designs and scale development
- Built and managed monitoring databases and systems; software development
- Evaluation consulting and support for project teams

EDUCATION

University of Basel

August 2023 – May 2026

Doctorate in Clinical Psychology & Epidemiology (Dr. rer. nat.)

Adaptation of deep-learning methods, LLMs, and agentic pipelines for the clinical context and their realistic validation beyond standard benchmarks; methodological focus.

- Trained custom models and fine-tuned LLMs; analyzed internal model representations via mechanistic interpretability
- Developed new validation frameworks – transferring methods from classical clinical prediction models and study designs to the LLM context, where established metrics are still lacking
- Intersection of clinical psychology, epidemiology, methodology, and statistics; applied e.g. in AI-assisted suicide prevention

University of Bern

August 2022 – September 2023

CAS Machine Learning

- Contemporary ML methods with a focus on graph networks, deep learning, and natural language processing

International Psychoanalytic University, Berlin

October 2019 – September 2021

M.Sc. Psychology (English Track)

- Final grade 1.2 (German scale); clinical psychology with a focus on statistics and methodology
- Thesis: "Personality, Parenting and Social Class: Transactional Effects Across Generations" (grade 1.0)

MSB Medical School Berlin

October 2015 – April 2019

B.Sc. Psychology

- Final grade 1.6 (German scale); focus on educational psychology and clinical diagnostics

CLINICAL TRAINING (ONGOING)

Berlin Academy of Psychotherapy (BAP)

Since 2025

Additional licensure CBT (children & adolescents)

Berlin Training Academy (BfA)

Since 2024

Integrative Behavioral Therapy

SKILLS

Programming Languages: Python, R, SQL, Kotlin, TypeScript, JavaScript, HTML, CSS

ML & DL Frameworks: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, Hugging Face Transformers, Sentence-Transformers; vLLM, Ollama

ML Methods: LLMs, Graph Neural Networks / deep graph learning, transformers & time-series transformers, agentic AI, neurosymbolic AI, RAG / Graph-RAG, recommender & matching systems, instruction fine-tuning, retraining & recalibration, mechanistic interpretability, explainable AI, NLP

MLOps & Infrastructure: Docker, Kubernetes, OpenShift, NixOS, Terraform, Google Cloud (GCP), CI/CD & build/deploy pipelines, DEB/RPM packaging, MLflow, Weights & Biases, observability, incident management

Data Engineering: ETL & data warehousing, ingestion & scraping pipelines, OCR & chunking pipelines, data standardization

Knowledge Graphs: Neo4j, Cypher, graph-based memory systems

LLM Providers & Tooling: IBM Watson, Azure, Ollama, vLLM, retrieval augmentation, prompt engineering

Methodology & Validation: Validation frameworks for ML/LLMs, clinical prediction models, study design, statistics, epidemiology

Domain: Clinical psychological diagnostics, mental-health AI, GDPR-compliant systems

LANGUAGES

German: native

English: fluent

Russian: advanced

SELECTED PUBLICATIONS

1. **Thomas, J.**, Lucht, A., Segler, J., et al. (2025). An Explainable Artificial Intelligence Text Classifier for Suicidality Prediction in Youth Crisis Text Line Users: Development and Validation Study. *JMIR Public Health and Surveillance*.
2. Rarey, F., **Thomas, J.**, Berghöfer, A., et al. (2024). The Association of Socioeconomic Status with the Success of Chat-Based Online Counseling for Children and Youth: A Latent Change Score Modeling Approach. *Internet Interventions*.
3. Efe, Z., Baldofski, S., Kohls, E., et al. (2023). Linguistic Variables and Gender Differences Within a Messenger-Based Psychosocial Chat Counseling Service for Children and Adolescents: Cross-Sectional Study. *JMIR Formative Research*.
4. Efe, Z., Baldofski, S., Eckert, M., et al. (2023). Who Are Frequent Chatters? Characterization of Frequent Users in a 24/7 Messenger-Based Psychological Chat Counseling Service for Children and Adolescents. *Internet Interventions*.
5. Baldofski, S., Kohls, E., Efe, Z., et al. (2022). Do Children and Young Adults Using a 24/7 Messenger-Based Psychosocial Chat Counselling Service Benefit in Terms of Seeking Further Help? *JMIR Mental Health*.
6. Kohls, E., Guenther, L., Baldofski, S., et al. (2022). Suicidal Ideation Among Children and Young Adults in a 24/7 Messenger-Based Psychological Chat Counseling Service. *Frontiers in Psychiatry*.
7. Eckert, M., Efe, Z., Guenther, L., et al. (2022). Acceptability and Feasibility of a Messenger-Based Psychological Chat Counselling Service for Children and Young Adults ("krisenchat"): A Cross-Sectional Study. *Internet Interventions*.